

Market-based Instruments

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Learning Objectives

After taking this class, you will be able to:

- *understand* the rationale of market instruments for solving environmental issues
- *understand* how water markets work in different countries
- *acknowledge* the potential challenges of market instruments

Can you recall the limitation of command-and-control regulation?

Recap: Compliance and beyond-compliance

- Command-and-control regulation offers **no incentive to improve the quality of the environment beyond the standard** set by a particular law. Once the command-and-control regulation has been satisfied, polluters have zero incentive to do better.
- command-and-control regulation is **inflexible**. It usually requires the **same standard for all polluters**, and often the same pollution-control technology as well.

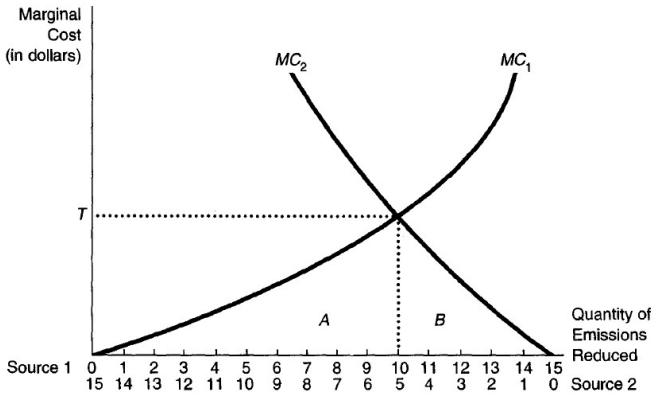
Can you explain why conventional environmental regulation is called as command-and-control approach?

Basic Rational of Market-based Instruments

Basic Rationale of Market-based Instruments

- Cap and trade system (tradable permits system) :
  - Cap: set up the maximum amount of emission or resource extraction
  - Trade: sell or issue a permit for polluters or resource extractors
  - Overall: A policy approach in which **use of the resource** is capped, **use permits** are allocated, and a **permit trading mechanism** is established.
- Any company that has high abatement costs can buy permits from another that has low costs, so reducing the total cost of abating pollution → ***Efficiency achieved!***
- Controlling the total weight of emissions in a manner that minimizes the costs of control

Cost-Effective Allocation of A Uniformly Mixed Pollutant

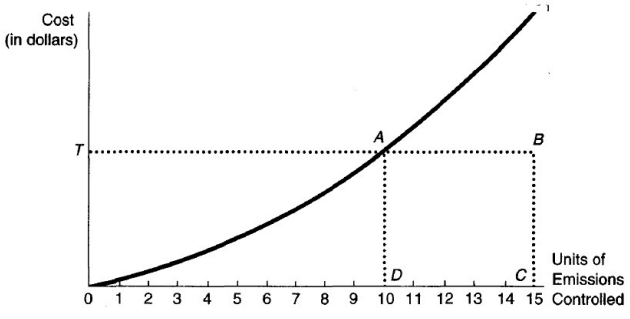


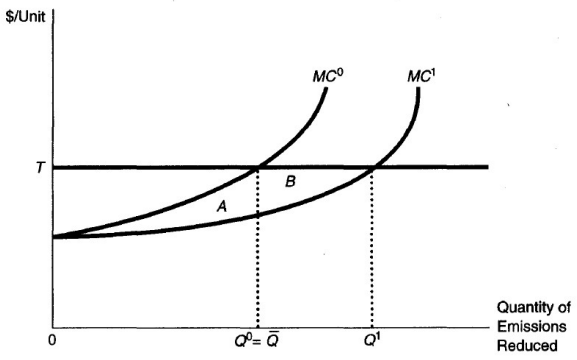
(Tietenberg, 2000)

- Uniformly mixed pollutant: insensitive to where the emissions are injected into the atmosphere
- A pollutant quickly becomes dispersed to the point where its spatial distribution is uniform (e.g. carbon)
- The cost of achieving a given reduction in emissions will be minimized if and only if the marginal costs of control are equalized for all emitters.

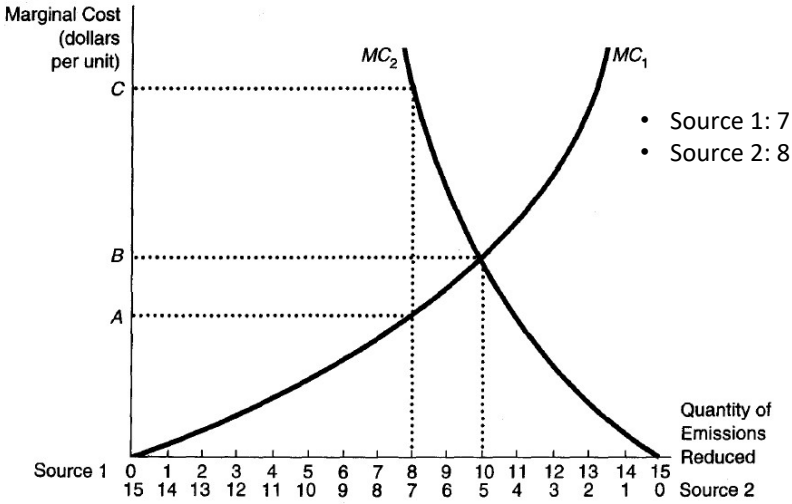
Cost-Effective Allocation of A Uniformly Mixed Pollutant

Emission Charges





Cost-Effective Allocation of A Uniformly Mixed Pollutant



- Source 1: 7 permits
- Source 2: 8 permits

Key Elements of Cap-and-Trade System

- Determining how much of the resource can be used (total allocable catch, emissions, water diversions) over what time frame, and with what exceptions for unusual conditions (droughts, hurricanes)
- Assign initial allocations
- Set the rules for transfer
- Monitor and enforce the activities

An Example: Water Rights Markets and Trading

What is water right?

- A ‘water right’ is the right to use water according to water quotas.
- Water rights are usually allocated by governments via Water Use Permits and can be 'traded' between users.
- Reforms let markets decide allocation & treat water rights as property rights to protect public interests.
- Water market functions as a “cap-and-trade” market.

Water Rights in Murray-Darling Basin (Australia)

- Before 1980s, water rights in Australia are tied to land rights.
- Water users can buy or sell their water rights, on a permanent or temporary basis.
- This water is traded on markets – within catchments, between catchments (where possible) or along river systems.



Water Rights in Murray-Darling Basin (Australia)



- How are the water rights allocated in the first place?

❖ Key Institutions

- **Murray–Darling Basin Agreement** (1989)**:** set out the principles of water sharing
- Water Act (2007): formalize the agreement
- Basin Plan: sets triggers for moving between water sharing tiers

❖ Key Actors

- **Murray-Darling Basin Authority (MDBA)**: an *independent* body to make sure each state gets the water it is entitled to from the major River Murray storages (former Murray-Darling Basin Commission)
- Three states: New South Wales, South Australia and Victoria
- Commonwealth Environmental Water Holder
- Australian Competition and Consumer Commission
- Bureau of Meteorology

Water Rights in Murray-Darling Basin (Australia)

- How are the water rights allocated in the first place?



South Australia entitlement ← supplied equally by ← NSW, Victoria

Inflows to Menindee Lake ← shared 50:50 (*when lakes are operated by MDBA) ← NSW, Victoria

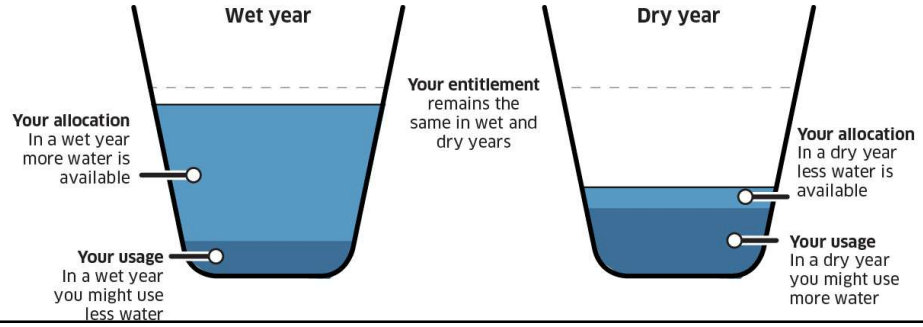
NSW tributary inflows downstream of Albury = 100% NSW water

Inflows upstream of Albury ← shared 50:50 ← NSW, Victoria

Victoria tributary inflows downstream of Albury = 00% Victoria water

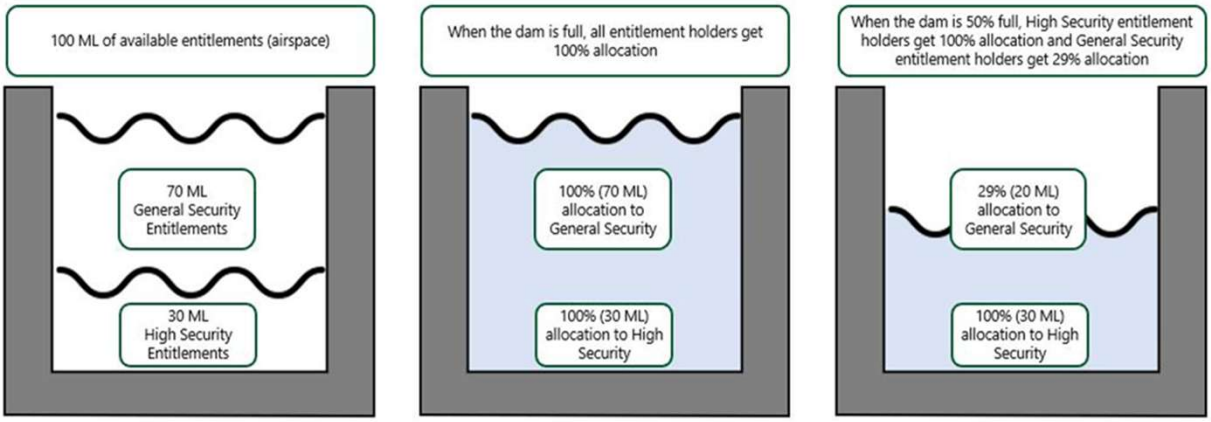
Water Rights in Murray-Darling Basin (Australia)

- What kind of water rights can be traded?
  - **Water entitlements:** rights to an ongoing share of water within a system
  - **Water allocations:** the amount of water distributed to users (water entitlement holders*) in a given year*
  - **Water usage:** how much water is actually used from the water that is allocated. An entitlement holder can continue to receive additional water until they have received a 100% allocation (their 'full entitlement').



Water Rights in Murray-Darling Basin (Australia)

- What kind of water rights can be traded?

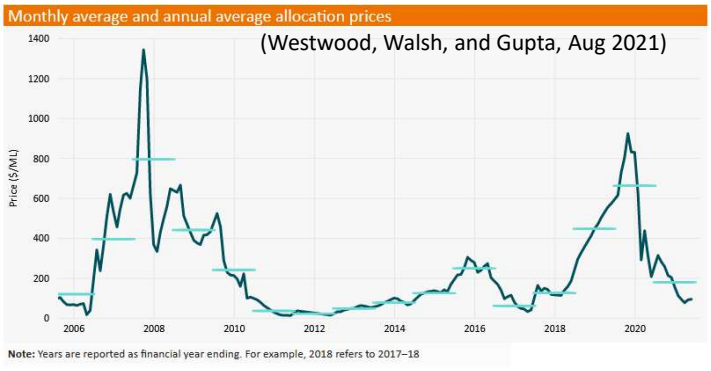


Water Rights in Murray-Darling Basin (Australia)

- How does water trading work?
  - Transparent information:
    - The Basin states and the Australian Government must make water announcements generally available.
    - People who sell their water must declare their sale price to their approval authority when they trade.
    - notify the parties involved in a trade if it is restricted or refused, while also listing the reasons for the decision
    - disclose any legal, commercial or equitable interest they have in a trade to those involved in the trade when processing it

Water Rights in Murray-Darling Basin (Australia)

- How does water trading work?
  - Water prices
    - Short term: climatic conditions, storage levels, irrigated agricultural commodity prices
    - Long term: agricultural productivity growth; the transition of farming businesses from low-value to high-value production



Water Rights in Murray-Darling Basin (Australia)

- Does it work?
  - helped deliver improved environmental outcomes
  - assisted irrigators' adaptation responses to climate risks, such as drought
  - increased the gross valued added of farming;
  - been regulated in ways to meet social goals

(Grafton, Horne, and Wheeler, 2016)

Water Market in the Colorado River Basin (the US)

- Water Bank (generally short-term or one-time sales)
  - A buyer who anticipates future water shortages can purchase the "option" to draw a certain amount of water from the seller at a future time.
  - An action group or board purchases water for environmental uses to restore stream flow or water level in wetlands

How to operate a water bank?

- Land fallowing: farmers agree to leave fallow some fields and sell the water they would normally use to irrigate those fields to other users, usually urban water districts.
- Water conservation: A metropolitan area might pay for water-saving irrigation and pumping infrastructure with an agreement that any water saved would be sold back to the city
- Recycling Wastewater
- Water storage: A user will store unused water in underground aquifers during wet years and pump this water out during shortfalls.

Water Rights Reform in China

- Water rights reform was initiated by the Ministry of Water Resources since July 2014. (*but several local water rights trade initiatives have been implemented since early 2000s.*)
- National level water rights reform: Ningxia, Jiangxi, Hubei, Inner Mongolia, Henan, Gansu, Guangdong
- Pilots at provincial level: Hebei, Xinjiang, Shandong, Shanxi, Shaanxi, Zhejiang
- China Water Exchange, the first national level platform for water rights trading

The Problems of Market Instruments

Perfect Market Conditions

- No public goods
- No externalities
- No monopoly buyers or sellers
- No increasing returns to scale
- No information problems
- No transaction costs

⋮

Cap-and-trade system has its advantages, but it is still a 'market'. The outcomes will be efficient only if certain conditions are met.

So, policy design matters!

Some Critiques to EU Emission Trading System

- Fell short of emission targets in the first two phases
- weak emissions caps; over allocation of free permits to participating firms
- volatility in emissions allowance prices: drop in carbon price to nearly zero in Dec, 2007
- Passing on the abatement task to the developing world
- Science and measurement issue

References

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- Tietenberg, T. 2000, Chapter 15, Economics of Pollution Control, in Environmental and Resource Economics, 340-355.